Comparing Correlations of U.S. Internet Growth Factors: Software Companies and the Economy

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Abstract

This study explores the intertwined relationship between the growth of American name-brand software companies, the U.S. Economy and the growth of Internet usage between 1990 and 2020. Through thorough data collection and correlational analyses, the research yields that, while there exists a strong correlation between the success of software companies and the increased popularity of the Internet in the United States, the same cannot be said for the broader economic indicators that were examined. These findings essentially challenge the assumption that economic growth is the primary stimulant for the expansion of the Internet, suggesting instead that technological advancements and software companies play a pivotal role. The implications of this idea extend to offer strategic insights for improving Internet access globally, emphasizing the importance of technological development alongside economic growth initiatives.

Literature Review

Introduction

Over the course of the last 40 years, the popularity and spread of the Internet in the United States has reached astonishing heights. In fact, it is estimated that 92% of the 2022 United States population used the Internet in some manner on a daily basis (The World Bank, 2021). That percentage has only increased since then, with estimates for its growth being placed at around 1.5 million new users compared to last year (Kemp, 2023). This significant growth has often been attributed to the success of name-brand, or globally successful, U.S. software companies.

The Origins of the Internet

The creation of the Internet is a story rooted in the collaborative efforts of numerous scientists, engineers, and visionaries over several decades. Its origins date back to the 1960s Cold War era when the United States Department of Defense sought a durable communication system that could survive nuclear warfare (Inria, 2019). Thus, the Advanced Research Projects Agency (ARPA) established ARPANET in 1969, connecting computers at Stanford University and the University of California, Los Angeles as the first secure wireless communication network. This development laid the foundation as the precursor for the modern Internet.

Consumerism of Name-Brand Companies

Throughout the 1980s and 1990s, the Internet expanded rapidly, transitioning from a military tool to a global network for research, education, and commercial use. In 1989, Tim Berners-Lee's creation of the World Wide Web reimagined how information was distributed, making the Internet accessible and user-friendly for people worldwide (Pew Research Center, 2014).

As soon as this access was brought to consumer discretion, early American software companies began designing programs and products that would garner the public appeal of the Internet. These were the companies that developed the first microprocessors, applications, and programs that are now crucial components of every working computer. Since then, these American software companies have grown tremendously in value, reach, and popularity that they are now considered “name-brand” (Yrjönkoski, 2019).

Name-Brand Software Companies

Some of the most known name-brand software companies of the current era include Microsoft, Apple, and Oracle. These three companies were established around the late 1980s and began to grow in value with the onset of the World Wide Web in the 1990s. Through their

contributions and development of products relating to the Internet, they have since achieved immense sales, profits, recognition, and stock market value. And now, Microsoft, Apple, and Oracle sit atop the top 5 largest software companies by market value (The Software Report, 2022).

Due to the popularity of these name-brand software companies, the public views their success as a major stimulator of the growth of Internet usage in the United States. However, while the successful business models, innovations, and social trends established by these large software companies may have contributed to this perception, the growth of the U.S. economy may have played an equal or even larger role in the growth of the Internet in the United States.

Relating Software Company Growth and Technology Adoption

This perception of software companies' growth as a contributor to increased Internet usage is supported by the idea that newer and more improved technologies develop as software companies grow. With better technologies entering the market, the overall consumerism of technology also increases. With increased technological consumerism, there is increased usage of the Internet across the nation. Hence, software companies' growth is associated with increased Internet usage.

One recent study supporting this presumption was conducted by Dr. Debra Hooks, a Data Science and Business Technology Program Manager at CitiBank, taking place from 2010-2019. Her evaluation followed several software companies and the populations of 15 separate countries to evaluate the extent to which market competition influences increased technological consumption.

Dr. Hooks' evaluation was accomplished through the use of a Predictive Panel Model and the analysis of software companies' business capabilities with a specific interest in the market

competition. This emphasis revealed that these software companies' competition boosts their individual marketing, administrative measures, product development, and stock market success.

After analyzing all these subfactors and determining that each influences increased technological consumerism and adoption, Dr. Hooks was then able to generalize that software companies' "...competitiveness is impactful to technology adoption because it increases wealth and economic stability in a country and allows for more technology to be utilized" (Hooks et al. 2022).

This conclusion effectively demonstrates a logical relationship between software companies' competition and the adoption of technologies. Dr. Hooks essentially reasons that increased market competition leads to the growth of these companies, as represented by boosted marketing, administrative measures, product development, and stock market success. These subfactors result in increased consumerism of technology, which in turn contributes to increased technological adoption.

Since increased technological adoption corresponds with increased Internet usage, the existence of this relationship expands the current body of knowledge, suggesting that there is indeed a correlation - one that can be identified and not only presumed - between software companies' growth and the growth of the Internet.

Defining the Growth of the U.S. Economy

A crucial element in fueling economic growth, economic development opens job opportunities and facilitates an improved quality of life for citizens. It refers to the production of wealth from which the nation gains in equal parts (Vaidya, 2023). Its measures include parameters such as Gross Domestic Product (GDP), Median Household Income, Trade Balance, and Unemployment Rate.

Relating Economic Growth and Technology Adoption

The main reason for the acknowledgment of America's economic growth as a contributor to increased Internet usage is centered around the idea that, as a population grows wealthier over time, consumerism of technology also increases. With increased technological consumerism, there is increased usage of the Internet across the nation. Hence, the logical flow presents that increased economic conditions ultimately contribute to increased Internet usage.

There have been several studies performed on the relationship between income and technology consumerism, one example being Sajda Qureshi's evaluation of economic growth on increased technology adoption. In this study, Qureshi, a professor of Information Systems at the University of Nebraska, analyzes the effects of per capita GDP growth on Information and Communication Technology (ICT) usage in 32 island states over the years 2009-2012.

In Qureshi's research with Lotfollah Najjar for the Routledge Taylor and Francis Group, the variable of average ICT is quantified by three indicators of usage: "...the number of mobile cellular telephone subscriptions (MCT) per 100 inhabitants; individuals using the Internet and Fixed (wired) Internet subscriptions per 100 inhabitants (INTU); and Broadband Internet subscriptions (BIS) per 100 inhabitants" (Qureshi et al. 2017). The variable of GDP growth consists of each island state's reported per capita Gross Domestic Profit, or the value created through goods and services.

The results of Qureshi and Najjar's study found that ICT growth rose 1.0% for every 0.6% increase in per capita GDP over the course of the 4 years the study was conducted. This essentially indicates that increased GDP does indeed lead to increased technology usage in populated areas. Thus, this finding expands the current body of knowledge by establishing that,

if the economy grows and improves, then the usage of technological systems that use the Internet will also increase.

That said, this study was conducted using data from island states, most of which have developing economies and fast-paced population growth. In their conclusion, Qureshi and Najjar reveal that significant population growth can twist the comparison of ICT growth percentage and GDP growth percentage since there are more people per capita (hence decreasing the number of ICTs per 100 people). For this reason, they state that the relationship between ICT and GDP growth may be indirect or inverse depending on the rate of population growth of the area being evaluated. However, their comparison of 0.6% GDP growth to 1.0% ICT growth is adjusted to account for areas of stable population growth, like the United States, so it can still be reasoned that economic growth contributes to increased Internet usage.

Implication of This Study

Though both the growth of software companies and the growth of the economy are logical contributors to the growth of the Internet, there has been no comparison of which factor has been the greater contributor to the popularity of the Internet in the United States. The reason for this gap stems from the public perception that the growth of the Internet has come due to the success of name-brand software companies.

Although Qureshi and Najjar's study complements the line of reasoning that the improvement of the economy is a contributor to increased Internet usage, their research focuses primarily on the growth of per capita GDP and its effect on increasing Internet Communications Technology usage. In other words, their findings cannot fully be generalized to reveal the extent to which the growth of the economy relates to the growth of Internet usage.

Furthermore, Qureshi and Najjar's study was conducted using populations from island states, which have small populations, volatile economies, and rapid population growth. These traits make it so that the findings of their study cannot be generalized to apply to much larger populations and more stable economies, like those of countries like the United States. This is yet another reason why explicit research must be conducted evaluating the relationship between the growth of the economy and the growth of Internet usage within the United States.

To compare the influences of software companies and the U.S. economy on Internet usage, this study will evaluate whether the stock growth of name-brand software companies or the growth of the U.S. economy is more closely correlated to the popularity of American Internet usage between 1990-2020. This analysis is hypothesized to yield that the U.S. economy is more related to Internet usage than is the growth of software companies, thus defying the assumption that American name-brand software companies made the Internet popular between 1990-2020.

Method

Collecting United States Internet Usage Data

The first step to collecting data for the United States' annual Internet Usage was to head to The World Bank. This website contains a large line graph in the middle of the page with a rectangular data chart right below it. For the sake of this study, only the line graph was utilized for data collection.

To collect accurate data for this study, the blue filters "Individuals using the Internet (% of population)" and "United States" above the line graph were checked off. These are crucial filters that were needed to streamline the line graph.

Next, the interactive line graph was hovered over to view the different data points for each year. When done so, a gray box containing a number and the date appeared. The number

displayed here represented the percentage of Americans using the Internet for that year. These are the data values that were then recorded for this study.

To store these data values, a Google Spreadsheet was created that is titled "By Year: Economic Factors vs. Internet Usage." The first column of this spreadsheet was then labeled "Year." The second column was labeled "Internet Usage (% of U.S. Population)."

After labeling properly, all the corresponding Years and Internet Use Percentages were entered into the spreadsheet. This was done by hovering over each dot and recording the Year and Internet Use Percentage in the corresponding columns of the spreadsheet. This step was completed for each dot from 1990 to 2020 (inclusive). In the end, there were 2 columns of data with 31 values in each.

Collecting GDP Growth Data

The first step to collecting GDP Growth Data was to head to The World Bank. This directed to a website with a large line graph and a chart of countries below it. Only the line graph was valuable to this study.

To collect accurate data for this component, the filters "GDP Growth (annual %)" and "United States" (both of which are highlighted in blue) were checked off. These were crucial filters that were needed to streamline the line graph.

Next, the interactive line graph was hovered over to view the different data points for each year. When done so, a gray box containing a number and the date appeared. The number displayed here represented the GDP Growth Percentage for that year. These are the data values that were then recorded for this study.

To store these data values, the Google Spreadsheet titled “By Year: Economic Factors vs. Internet Usage” was opened up. The third column of this spreadsheet was then labeled “GDP Growth (%).”

After labeling properly, all the corresponding GDP Growth Percentages were entered into the spreadsheet. This was done by hovering over each dot and recording the GDP Growth Percentages per the corresponding rows of the Year column. This step was completed for each dot from 1990 to 2020 (inclusive). In the end, there were 31 values in the column.

Collecting Median Household Income Data

The first step to collecting Median Household Income data was going to The Federal Reserve Bank of St. Louis. This led to a website with a large line graph in the middle of it.

To collect accurate data for this component, the Date Filter was modified to show “1990-01-01” to “2020-01-01”. This filter made the graph more relevant to this study's timeline and improved the data collection's feasibility.

Next, the interactive line graph was hovered over to view the different data points for each year. When done so, a white box popped up by the mouse cursor detailing the Year and Median Household Income for that year. The median household incomes were the data values that were recorded for this study.

To store these data values, the Google Spreadsheet titled “By Year: Economic Factors vs. Internet Usage” was accessed. The fourth column of this spreadsheet was labeled “Median Household Income (USD).”

After labeling properly, all the corresponding Median Household Incomes were recorded into the spreadsheet. This was done by hovering over each dot and inputting the Median

Household Incomes per the corresponding rows of the Year column. This step was completed for each dot from 1990 to 2020 (inclusive). In the end, there were 31 values in the column.

Collecting Unemployment Data

The first step to collecting Unemployment Rate data was heading to Mactrotrends - U.S. Unemployment. This led to a website with a line graph, a two-way bar chart, and two large data charts. This study only used the United States Unemployment Rate data chart.

Scrolling down to the data chart of the U.S. Unemployment Rate yielded a page with several data columns. This study only used the data in the column titled Unemployment Rate (%).

To store these data values, the Google Spreadsheet titled “By Year: Economic Factors vs. Internet Usage” was opened. The fifth column of this spreadsheet was then labeled “Unemployment Rate (%).”

After labeling properly, all the corresponding Unemployment Rate Percentages were logged into the spreadsheet. This was done by recording the Unemployment Rates per the corresponding rows of the Year column. This step was completed for each dot from 1990 to 2020 (inclusive), with there being 31 values in the column.

Collecting Trade Balance Data

To collect U.S. Trade Balance data, the U.S. Census Bureau was accessed. This link directed to a website containing many charts of information about U.S. trade. Each of these charts contained Import, Export, and Trade Balance data for each year in the U.S. from 2023 to 1989, but only the column for Trade Balance was useful to this study. Specifically within this column, only the value in the last row (shaded in grey) was used.

To store these data values, the Google Spreadsheet titled "By Year: Economic Factors vs. Internet Usage" was accessed. The sixth column of this spreadsheet was then labeled "Trade Balance (millions USD)."

After labeling properly, the website was scrolled through to obtain and record the total Trade Balance for each chart from 1990 to 2020 (inclusive). This process started with the chart for 1990 and was completed for each chart value from 1990 to 2020 (inclusive). In the end, there were 31 values in the column.

Collecting Apple, Microsoft, and Oracle Stock Data

The first step to collecting Apple, Microsoft, and Oracle Stock Price data was heading to their respective Macrotrends websites. Heading to the sites yielded a webpage with a line graph, a two-way bar chart, and a large data chart. This study only utilized the data chart for each software company. This data chart was then scrolled down upon to access the data within the Average Stock Price column.

To store these data values, a Google Spreadsheet titled "By Year: Software Company Stock Prices vs. Internet Usage" was created. The first column of this spreadsheet was labeled "Year." The second column was labeled "Internet Usage (% of U.S. Population)." The third column was labeled "Apple - Average Stock Price (USD)." The fourth and fifth columns were respectively labeled "Microsoft - Average Stock Price (USD)" and "Oracle - Average Stock Price (USD)."

After labeling properly, the Year and Internet Usage values from the Google Spreadsheet "By Year: Economic Factors vs. Internet Usage" were copied and pasted into Columns 1 and 2 of the current spreadsheet. Then, for Columns 3, 4, and 5, all the corresponding Apple, Microsoft, and Oracle Stock Prices were entered into the spreadsheet. This was done by recording the Stock

Prices per the corresponding rows of the Year column. This step was completed for each chart value from 1990 to 2020 (inclusive). In the end, there were 5 columns of data with 31 values in each.

Statistical Tests

After recording these data, correlative analyses were conducted using each economic variable and each software variable, with the Internet Usage remaining the stable y-variable for each test. These data were also graphed using Google Spreadsheet features to display scatterplots and Least Squares Regression Lines for each correlation, along with the corresponding r and r^2 values. After analyzing the r and r^2 values, the strength of the correlation was then determined, revealing whether the economic or software company variables yield a closer relationship with the constant variable of Internet Usage.

Results

Table 1 - By Year: Economic Factors vs. Internet Usage

Year	Internet Usage (% of U.S. Population)	GDP Growth (%)	Median Household Income (USD)	Unemployment Rate (%)	Trade Balance (millions USD)
1990	1.00	1.9	61500	5.6	-101,029.50
1991	1.00	-0.1	59710	6.8	-66,249.80
1992	2.00	3.5	59210	7.5	-84,500.90
1993	2.00	2.8	58920	6.9	-115,568.00
1994	5.00	4	59550	6.12	-150,627.00
1995	9.00	2.7	61440	5.65	-158,801.00
1996	16.00	3.8	62350	5.45	-170,213.00
1997	22.00	4.4	63640	5	-180,522.00
1998	30.00	4.5	65980	4.51	-229,759.00
1999	36.00	4.8	67650	4.22	-328,820.00
2000	43.00	4.1	67470	3.99	-436,103.00
2001	49.00	1	66360	4.73	-411,899.00
2002	59.00	1.7	65820	5.78	-468,262.00
2003	62.00	2.8	65860	5.99	-532,350.00
2004	65.00	3.9	65760	5.53	-654,828.50
2005	68.00	3.5	66780	5.08	-772,374.20
2006	69.00	2.8	67520	4.62	-827,970.00
2007	75.00	2	68610	4.62	-808,765.00
2008	74.00	0.1	66280	5.78	-816,200.00
2009	71.00	-2.6	65850	9.25	-503,583.00
2010	72.00	2.7	64300	9.63	-635,365.00
2011	70.00	1.5	63350	8.95	-725,447.00
2012	75.00	2.3	63350	8.07	-730,446.30
2013	71.00	1.8	65740	7.37	-689,469.90
2014	73.00	2.3	64900	6.17	-734,482.30
2015	75.00	2.7	68410	5.28	-745,483.00
2016	86.00	1.7	70840	4.87	-735,326.00
2017	87.00	2.2	72090	4.36	-792,395.90
2018	88.00	2.9	73030	3.9	-870,358.40
2019	89.00	2.3	78250	3.67	-845,759.20
2020	91.00	-2.8	76660	8.05	-901,481.80

This table depicts the yearly fluctuations in U.S. economic conditions and the percentage of Americans with Internet Usage. The yearly progression is shaded in the violet data column, while the Internet Usage percentage values are shaded in turquoise. The category of U.S. economic conditions (shaded in blue) consists of 4 data columns: GDP Growth percentage, Median Household Income in USD, Unemployment Rate percentage, and Trade Balance in millions of USD. The column of GDP Growth is defined as the percentage change (as compared to the year prior) in U.S. Gross Domestic Product (GDP), or the total value of a country's goods and services for one year. This data was provided by The World Bank, which is a renowned publisher of economies and countries' growth and an organization that is regarded for using internationally approved standards to collect its data. The column of Median Household Income is defined as the center, or generalized value, of the annual income distribution consisting of all registered American census-takers. This data was provided by the U.S. Census Bureau, a subdivision of the U.S. Government, which every year sends out a mandatory economic census to all households in the United States and then publishes the results for public research. The column Unemployment Rate describes the percentage of the U.S. population unemployed, or without a job, for that year, the data of which was collected by The World Bank and displayed in the regulated economic database MacroTrends. The column of Trade Balance is defined as the difference between U.S. exports (goods shipped out of the country) and imports (goods shipped into the country) in millions of U.S. dollars. This data was also supplied by the U.S. Census Bureau, which annually updates, reviews, and republishes the Trade Balance dataset following the distribution of the U.S. Census. The division line above the data values for 2009 represents the economic regression of 2008, while the division line above the data values for 2020 represents the economic regression as a result of the 2020 COVID-19 Pandemic. The implications of these two events on the U.S. economic factors and Internet usage will be discussed later in this paper.

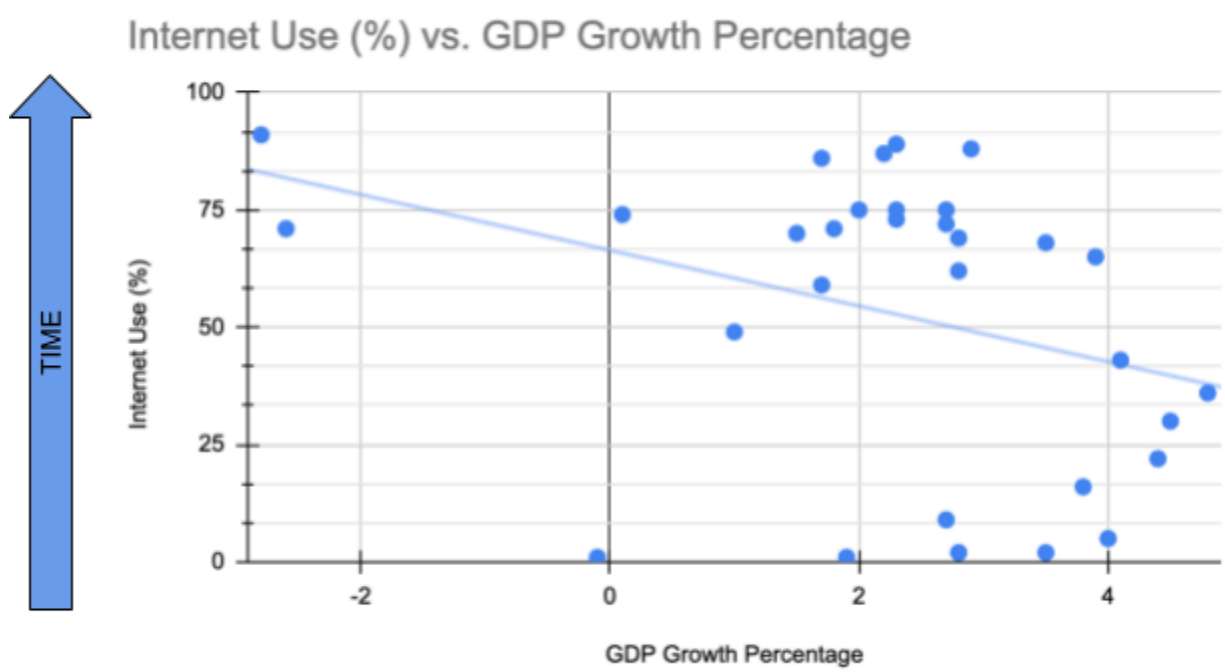
Table 2 - By Year: Software Company Stock Prices vs. Internet Usage

Year	Internet Usage (% of U.S. Population)	Apple - Average Stock Price (USD)	Microsoft - Average Stock Price (USD)	Oracle - Average Stock Price (USD)
1990	1.00	0.27	0.54	0.31
1991	1.00	0.38	1.01	0.21
1992	2.00	0.4	1.57	0.37
1993	2.00	0.3	1.63	0.96
1994	5.00	0.25	2	1.52
1995	9.00	0.31	3.24	2.27
1996	16.00	0.19	4.71	3.42
1997	22.00	0.14	9.44	4.29
1998	30.00	0.23	15.39	3.74
1999	36.00	0.44	27.34	8.65
2000	43.00	0.69	23.8	29.14
2001	49.00	0.31	19.53	14.50
2002	59.00	0.29	17.04	9.13
2003	62.00	0.28	16.37	10.05
2004	65.00	0.54	17.36	9.98
2005	68.00	1.41	18.3	10.56
2006	69.00	2.15	18.85	12.54
2007	75.00	3.89	22.13	16.06
2008	74.00	4.3	19.64	16.47
2009	71.00	4.45	17.36	16.63
2010	72.00	7.88	20.83	21.22
2011	70.00	11.03	20.54	26.33
2012	75.00	17.5	24.15	25.23
2013	71.00	14.65	27.14	28.83
2014	73.00	20.5	36.48	34.77
2015	75.00	27.13	41.21	35.80
2016	86.00	24.13	50.09	34.97
2017	87.00	35.39	66.88	42.13
2018	88.00	45.12	95.67	44.15
2019	89.00	50.49	125.31	50.16
2020	91.00	93.52	187.41	52.38

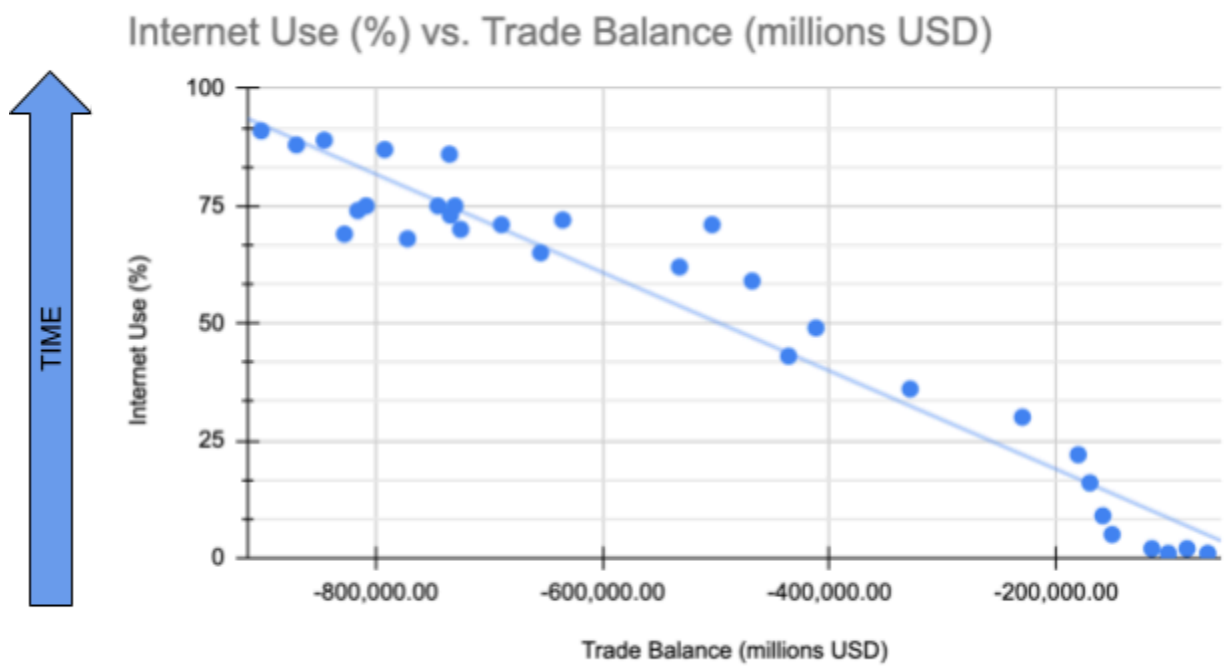
This table depicts the yearly fluctuations in stock value of the 3 large name-brand software companies and the percentage of Americans with Internet Usage. The yearly progression is shaded in the violet data column, while the Internet Usage percentage values are shaded in turquoise. The category of U.S. name-brand software companies (shaded in blue) consists of 3 data columns: Apple's Average Stock Price in USD, Microsoft's Average Stock Price in USD, and Oracle's Average Stock Price in USD. Each data value found in these columns essentially represents the average price for 1 stock of the company for every day in a given year. The stock price data for each of these 3 columns was collected from MacroTrends, a regulated economic database that collects daily, monthly, and annual data on various components of the U.S. economy. Within MacroTrends, this specific data was taken from the stock market analysis page, and specifically, the Stock Price History section. This particular section displays charts and data tables of each company's average annual price per stock progression. The division line above the data values for 2009 represents the economic regression of 2008, while the division line above the data values for 2020 represents the economic regression as a result of the 2020 COVID-19 Pandemic. The implications of these two events on the name-brand software companies' stock prices and U.S. Internet usage will be discussed later in this paper.

Correlative Analyses

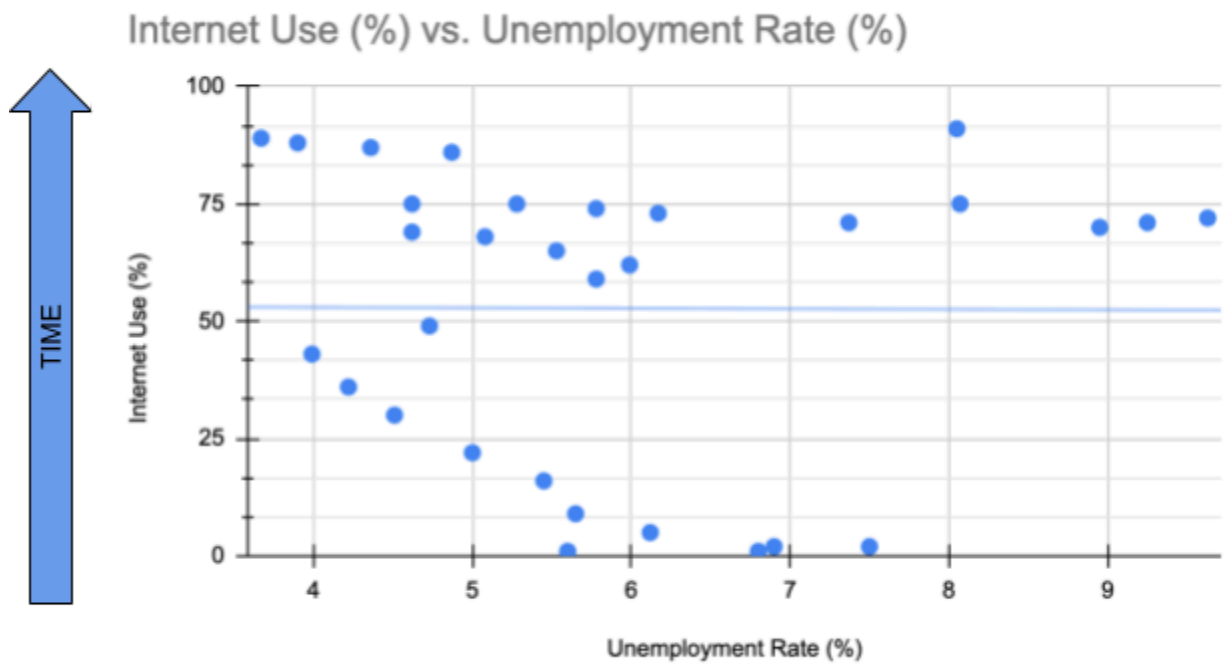
Question	X-axis Y-axis n	Hypothesis r-value r^2-value	Interpretation



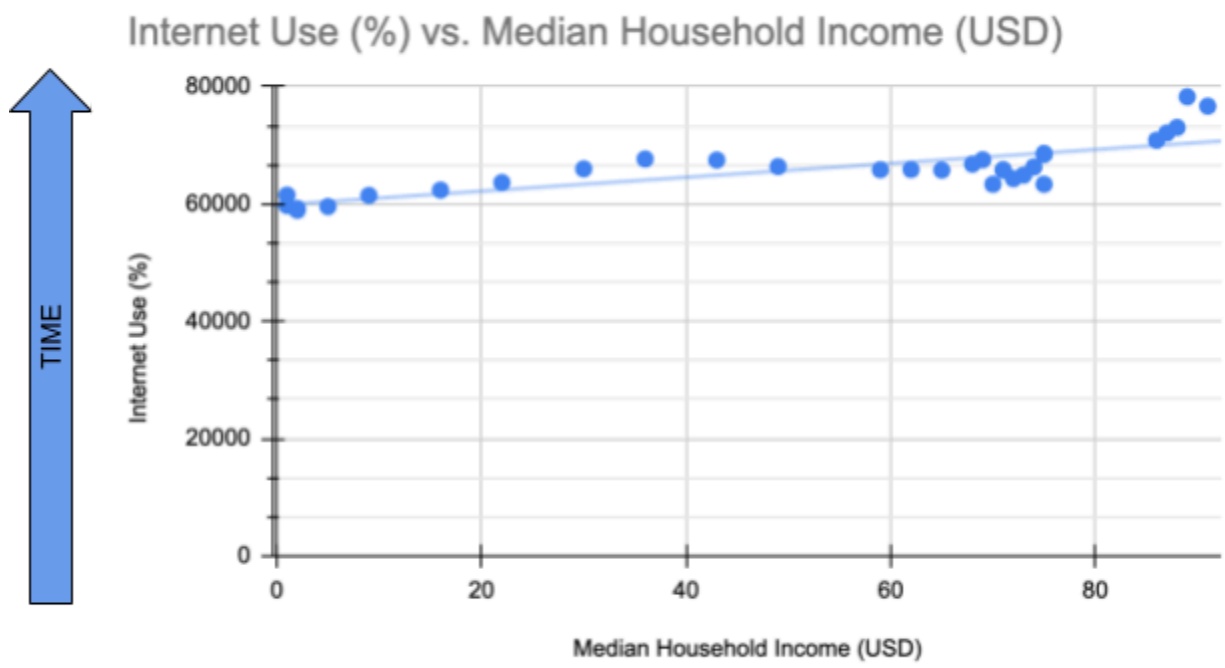
Is there a correlation between Internet Use (%) and GDP Growth Percentage?	X: GDP Growth Percentage Y: Internet Use (%) n = 31	$0.75 < r < 1$ $r = -0.3416$ $r^2 = 0.1167$	Although technically a negative correlation, the relationship between Internet Use (%) and GDP Growth Percentage is weak (the closer the r-value is to zero, the weaker the relationship). The r^2 value explains that approximately 11% of the variance of the Internet Use (%) being studied is explained by the variance of the GDP Growth Percentage.



Is there a correlation between Internet Use (%) and Trade Balance (millions USD)?	X: Trade Balance (millions USD) Y: Internet Use (%) n = 31	$-0.75 < r < -1$ $r = -0.9678$ $r^2 = 0.9366$	This is a strong negative correlation as given by the r-value being very close to -1, which means that high Internet Use (%) scores go with low Trade Balance scores (and vice versa). The r^2 value explains that approximately 94% of the variance of the Internet Use (%) being studied is explained by the variance of the Trade Balance (millions USD).

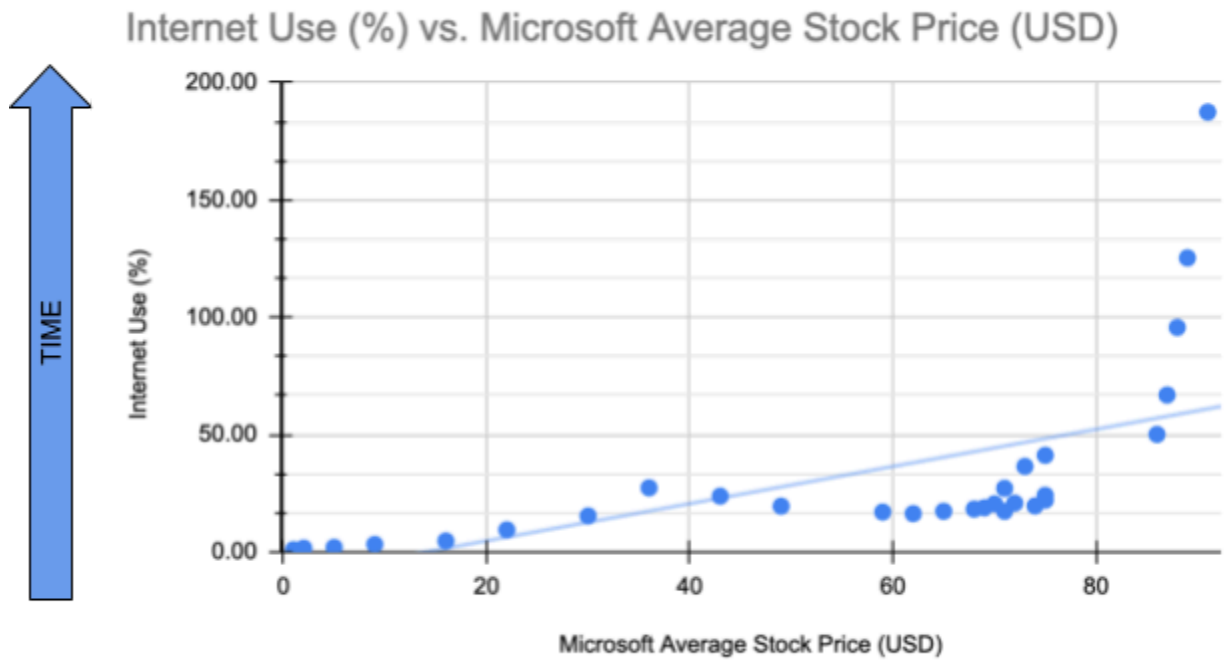


Is there a correlation between Internet Use (%) and Unemployment Rate (%)?	X: Median Household Income (USD) Y: Internet Use (%) n = 31	$0.75 < r < 1$ $r = -0.0054$ $r^2 = 0$	Although technically a negative correlation, the relationship between Internet Use (%) and Unemployment Rate (%) is essentially nonexistent (the nearer the r-value is to zero, the weaker the relationship). The r^2 value explains that the variance of the Internet Use (%) being studied cannot be explained by the variance of the Unemployment Rate (%).

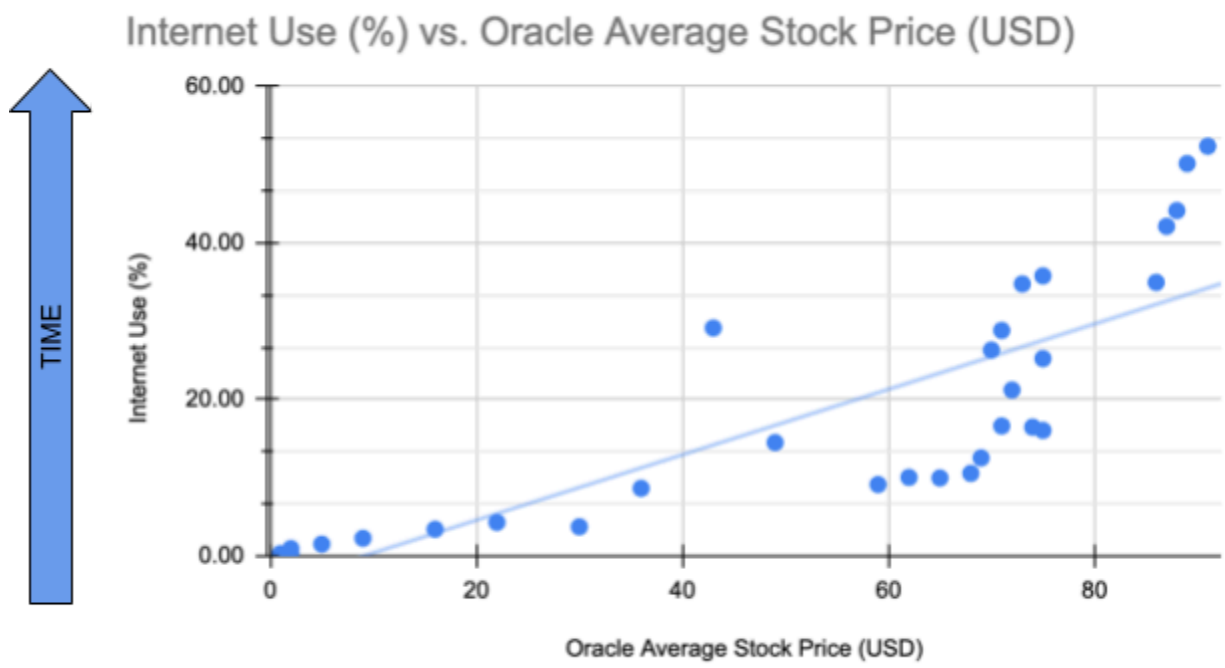


Is there a correlation between Internet Use (%) and Median Household Income (USD)?	X: Median Household Income (USD) Y: Internet Use (%) n = 31	$0.75 < r < 1$ $r = 0.7843$ $r^2 = 0.6151$	This is a strong positive correlation with the r-value being relatively close to 1, which means that high Internet Use (%) scores go with high Median Household Income scores (and vice versa). The r^2 value explains that approximately 62% of the variance of the Internet Use (%) being studied is explained by the variance of the Median Household Income (USD).

Is there a correlation between Internet Use (%) and Apple Average Stock Price (USD)?	X: Apple Average Stock Price (USD) Y: Internet Use (%) n = 31	$0.75 < r < 1$ $r = 0.588$ $r^2 = 0.3457$	This is a moderate positive correlation with the r-value being greater by 0.5 (though not by much), which means there is a tendency for high Internet Use (%) scores to go with high Apple Average Stock Price scores (and vice versa). The r^2 value explains that approximately 35% of the variance of the Internet Use (%) being studied is explained by the variance of the Apple Average Stock Price (USD).



Is there a correlation between Internet Use (%) and Microsoft Average Stock Price (USD)?	X: Microsoft Average Stock Price (USD) Y: Internet Use (%) n = 31	$0.75 < r < 1$ $r = 0.6139$ $r^2 = 0.3769$	This is a moderate positive correlation with the r-value being greater than 0.5 (though not by much), which means there is a tendency for high Internet Use (%) scores to go with high Microsoft Average Stock Price scores (and vice versa). The r^2 value explains that approximately 38% of the variance of the Internet Use (%) being studied is explained by the variance of the Microsoft Average Stock Price (USD).



Is there a correlation between Internet Use (%) and Oracle Average Stock Price (USD)?	X: Oracle Average Stock Price (USD) Y: Internet Use (%) n = 31	$0.75 < r < 1$ $r = 0.8174$ $r^2 = 0.6681$	This is a strong positive correlation with the r-value being relatively close to 1, which means that high Internet Use (%) scores go with high Oracle Average Stock Price scores (and vice versa). The r^2 value explains that approximately 67% of the variance of the Internet Use (%) being studied is explained by the variance of the Oracle Average Stock Price (USD).

Discussion Pt.1

Software Company Correlations

While this study centers around the popular assumption that the large U.S. software companies supported the immense growth of the Internet in the United States from 1990-2020, the researcher first finds it necessary to statistically test the validity of this preconditional opinion. The correlational analyses of the three most successful name-brand software companies

(Apple, Microsoft, Oracle) and their respective Average Stock Price (USD) with the constant variable of U.S. Internet Use (%) from 1990-2020 are thus conducted.

These tests each yield positive correlations between Average Stock Price (USD) and Internet Use (%), suggesting that as time increases and Average Stock Price increases (USD), Internet Use (%) also increases. The correlation coefficients (r-values) of the Apple and Microsoft stocks (0.5880 and 0.6139 respectively) depict moderately strong correlations between the software companies' stock growth and U.S. Internet growth, whereas the correlation coefficient of the Oracle stock (0.8174) depicts a very strong correlation between the one software company's stock growth and U.S. Internet growth.

Due to the strong and positive correlational nature of the software companies' Average Stock Price (USD) with the constant variable of U.S. Internet Use (%), it is concluded that there is a reliable relationship between the growth of name-brand software companies and the growth of the Internet in the United States. Thus, the assumption that name-brand U.S. software companies supported the growth of the Internet in the United States can be sustained.

Economic Correlations

To evaluate the strength of the relationship between the growth of the U.S. economy and the growth of U.S. Internet Use from 1990-2020, each of the four economic indicators (GDP Growth Percentage, Trade Balance, Median Household Income, Unemployment Rate) are tested in a correlational analysis with the constant y-variable of Internet Use (%). Based on the strength of the correlation (proximity of the correlation coefficient/r-value to -1 or 1), each of these factors is analyzed in support of or denial of the initial hypothesis.

The weak correlations of GDP Growth Percentage and Unemployment Rate with the constant variable of Internet Use (r-values of -0.3416 and -0.0054 respectively) present a

statistically insignificant and non-explanatory nature between the contributing variables. Due to the low r-values and their proximity to 0, these two correlations suggest that there is virtually no relationship between these economic variables and Internet Use between the years 1990-2020. These tests thus do not support the initial hypotheses of $0.75 < r < 1$ for the GDP correlation and $-1 < r < -0.75$ for the Unemployment correlation.

Conversely, the strong correlations of Trade Balance and Median Household Income with the constant variable of Internet Use (r-values of -0.9678 and 0.7843 respectively) depict a statistically significant relationship between the contributing variables. Due to the strength of the r-values (as in proximity to 1 or -1), these two correlations suggest that there is an explanatory relationship between these economic variables and Internet Use between the years 1990-2020. These tests thus support the initial hypotheses of $-1 < r < -0.75$ for the Trade Balance correlation and $0.75 < r < 1$ for the Median Household Income correlation.

After conducting these tests, it is found that the economic growth indicators (GDP Growth Percentage, Trade Balance, Median Household Income, Unemployment Rate) underscore a nuanced and nonlinear relationship between the growth of the economy and Internet Use in the United States. While two of the correlational analyses conducted elicited a statistically significant relationship between the economic variable and Internet Use, the other two analyses yielded weak and insignificant relationships between the contributing variables. This draws the conclusion that only certain indicators of U.S. economic growth hold a reliable relationship with Internet Use. Thus, the hypothesis of the growth of the U.S. economy having a strong relationship with the growth of the Internet in the United States from 1990-2020 cannot be supported.

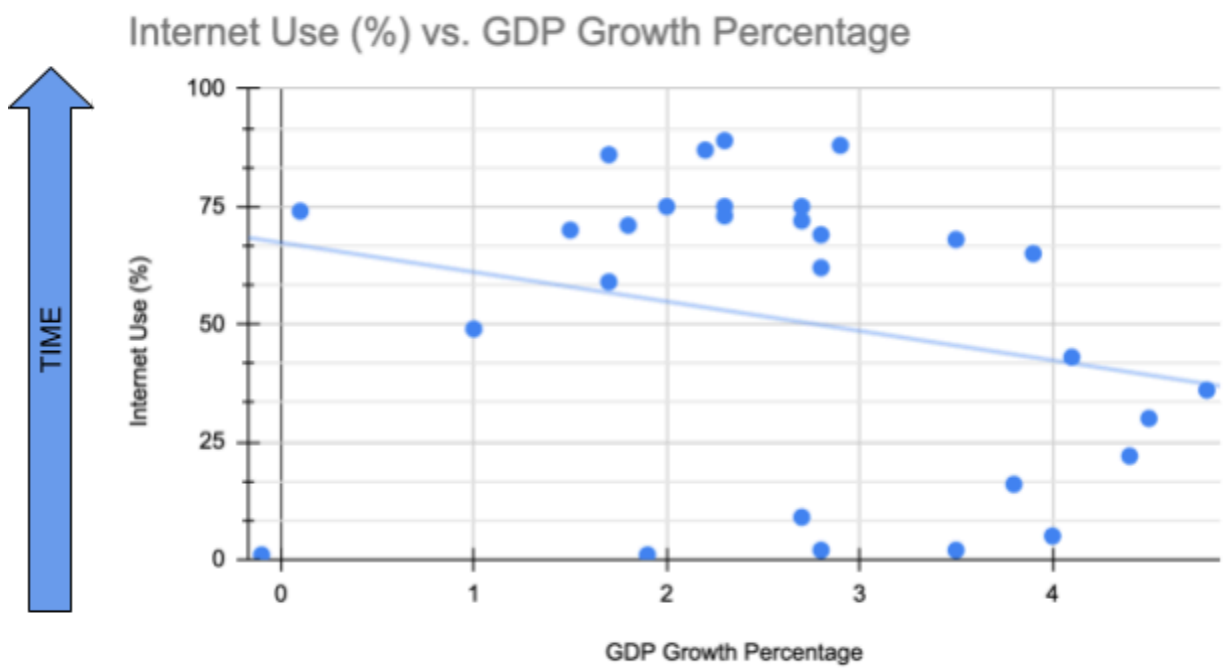
Limitations Pt.1

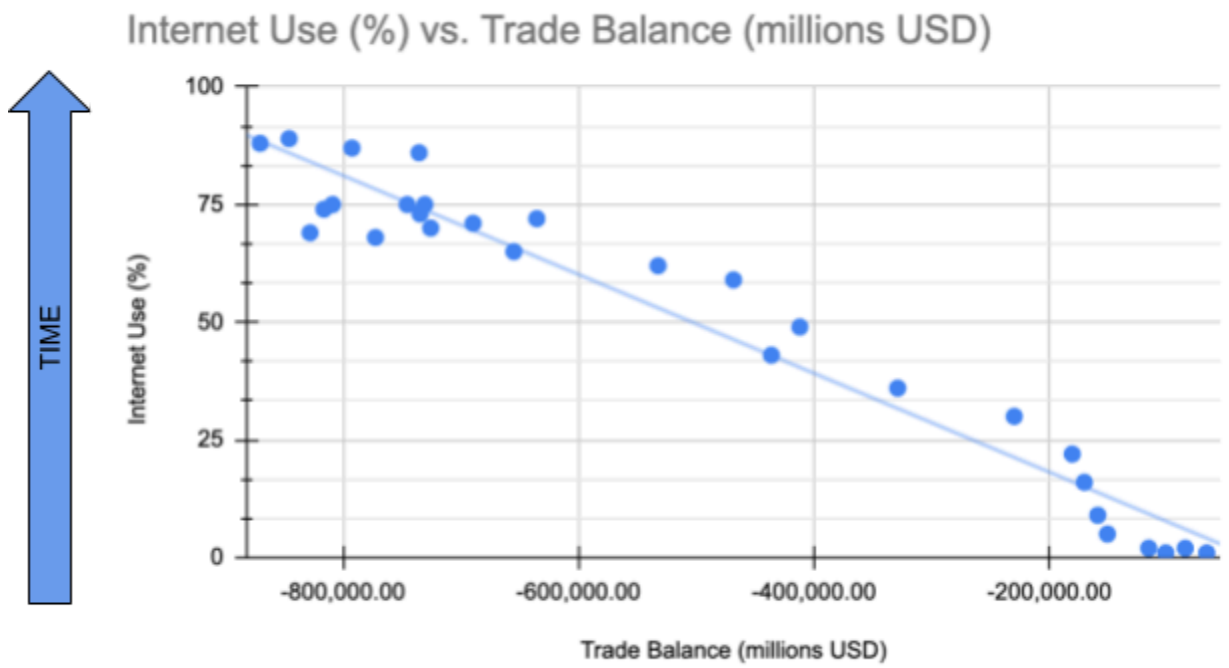
One limitation of this study is that each of the economic indicators and software stocks utilized in the correlational analyses is highly susceptible to economic changes such as market volatility, inflation, reduced consumerism, and decreased production. These changes occur during economic recessions, or periods of stagnated economic activity and depleted output.

Two such periods occurred during the scope of this study: the Great Recession of 2007-2009 and the COVID-19 Recession of 2020. The specific implications of these recessions included extremely high unemployment rates, large decreases in GDP growth percentage, vast income cuts, stock market volatility, and a significant reduction in global supply and demand—each of which impacted the economic indicators and software stock prices analyzed in this study.

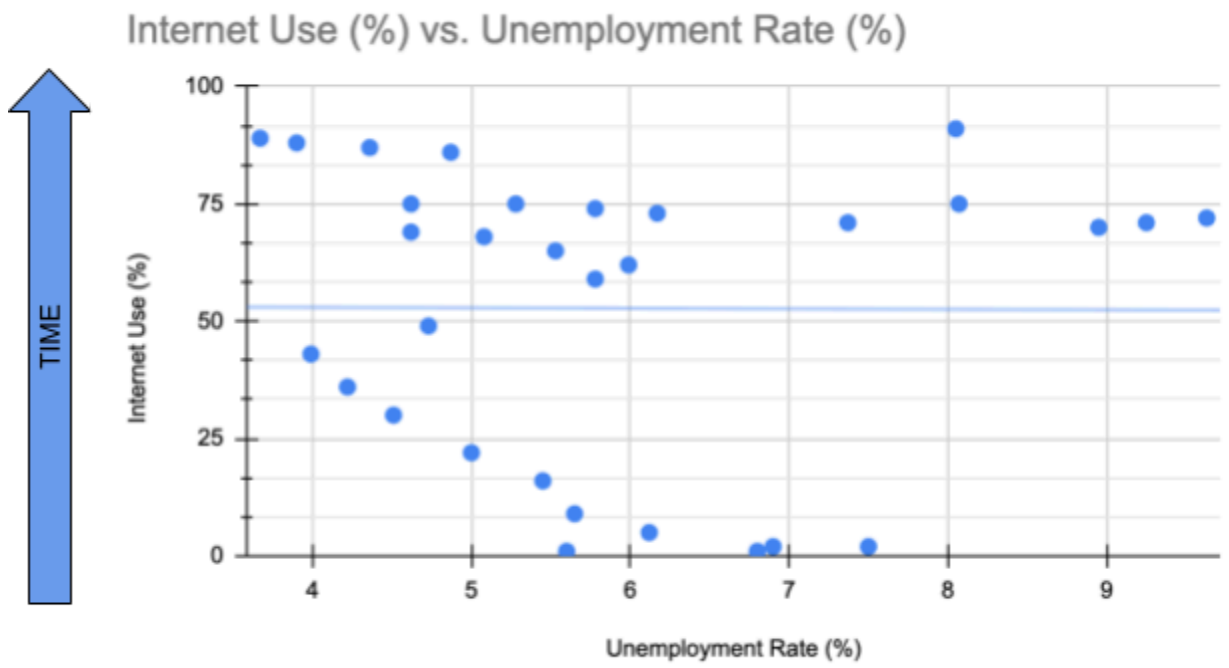
Hence, the limitation posed is the presence of data during the economic recession years in the overall dataset from 1990-2020. These two years hold characteristically lower data values than each of the remaining "normal" economic years, weakening the strength (r-value) and variance (r^2 value) of the correlations and skewing the line of best fit to overall lower data. For this reason, additional correlational analyses are conducted below with the omission of data values for the years 2008 and 2020.

Correlative Analyses - Recessions Removed

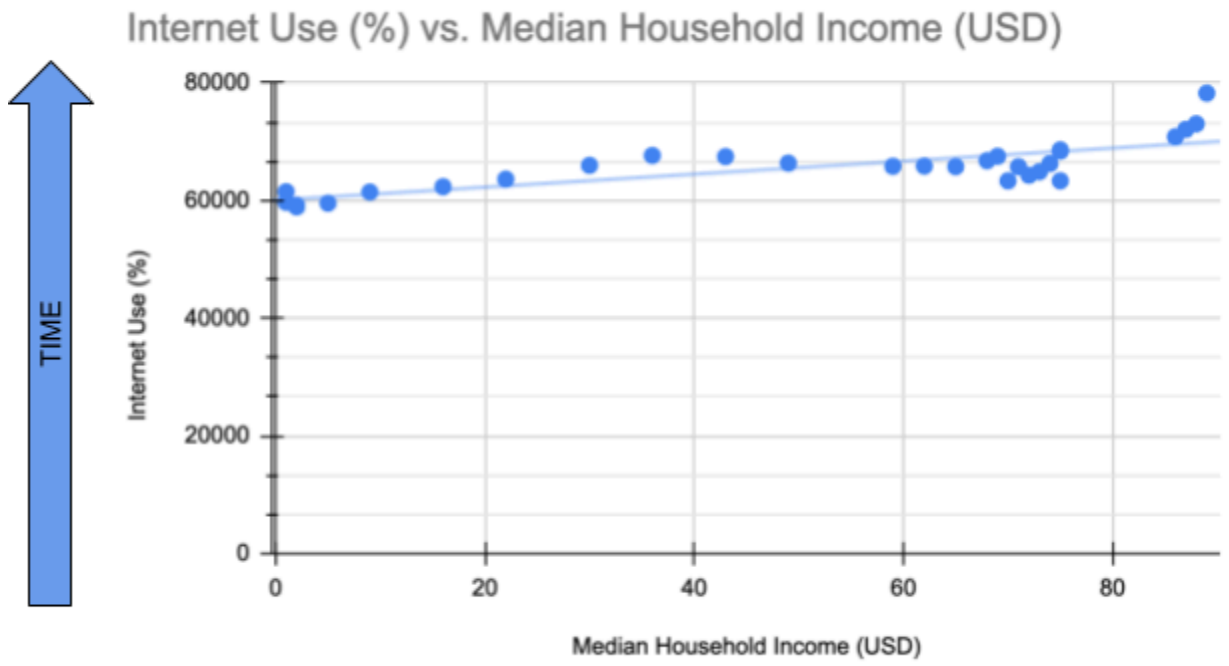
Question	X-axis Y-axis n	Hypothesis r Value r^2 Value	Interpretation
			
Is there a correlation between Internet Use (%) and GDP Growth Percentage?	X: GDP Growth Percentage Y: Internet Use (%) n = 29	$0.75 < r < 1$ r = -0.2466 $r^2 = 0.0608$	Although technically a negative correlation, the relationship between Internet Use (%) and GDP Growth Percentage is weak (the closer the r-value is to zero, the weaker the relationship). The r^2 value explains that approximately 6% of the variance of the Internet Use (%) being studied is explained by the variance of the GDP Growth Percentage.



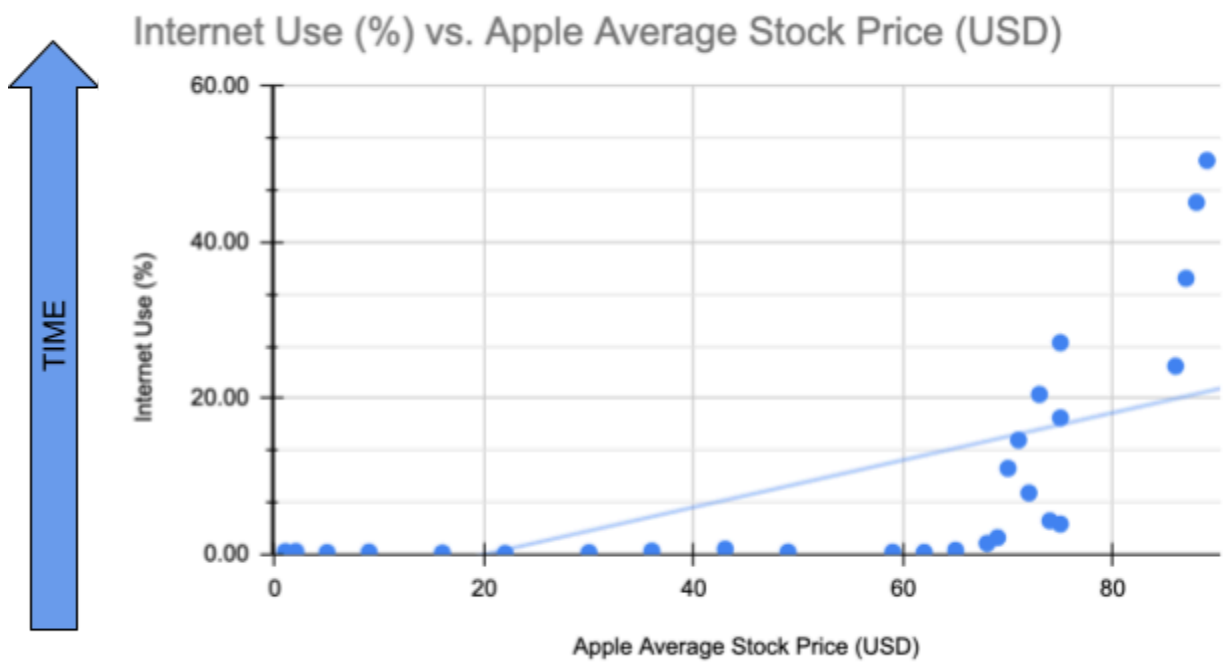
Is there a correlation between Internet Use (%) and Trade Balance (millions USD)?	X: Trade Balance (millions USD) Y: Internet Use (%) n = 29	$-0.75 < r < -1$ $r = -0.9737$ $r^2 = 0.9481$	This is a strong negative correlation as given by the r-value being very close to -1, which means that high Internet Use (%) scores go with low Trade Balance scores (and vice versa). The r^2 value explains that approximately 95% of the variance of the Internet Use (%) being studied is explained by the variance of the Trade Balance (millions USD).



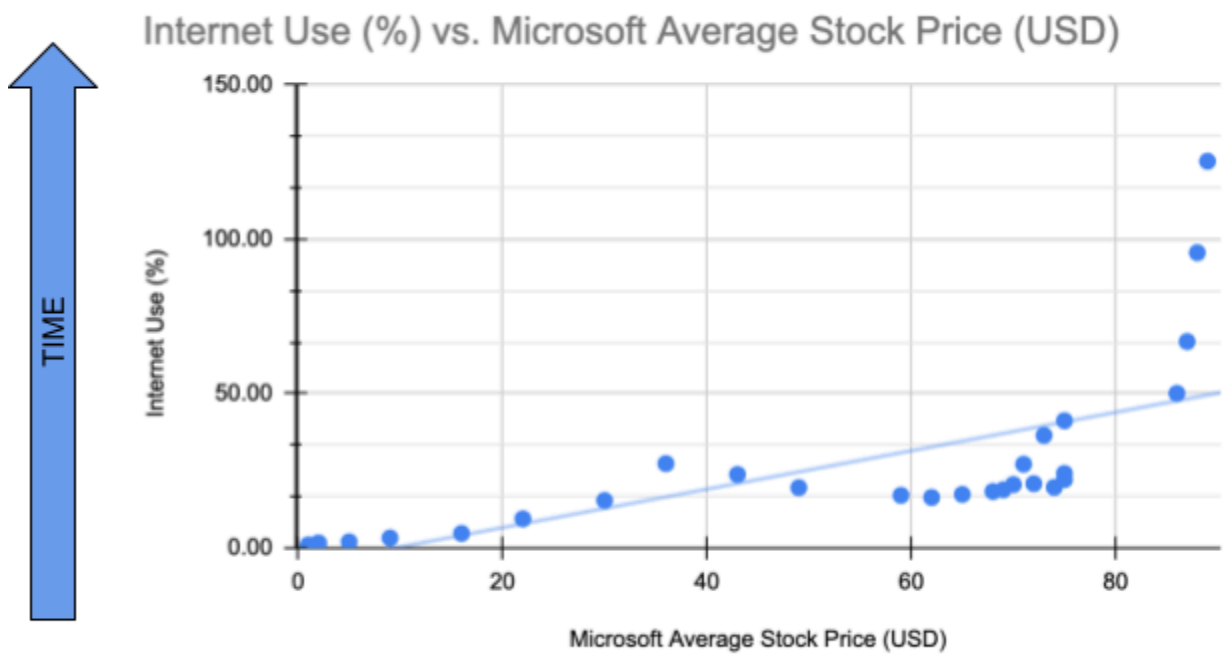
Is there a correlation between Internet Use (%) and Unemployment Rate (%)?	X: Median Household Income (USD) Y: Internet Use (%) n = 29	$0.75 < r < 1$ $r = -0.1255$ $r^2 = 0.0158$	Although technically a negative correlation, the relationship between Internet Use (%) and Unemployment Rate (%) is essentially nonexistent (the nearer the r-value is to zero, the weaker the relationship). The r^2 value explains that the variance of the Internet Use (%) being studied cannot be explained by the variance of the Unemployment Rate (%).



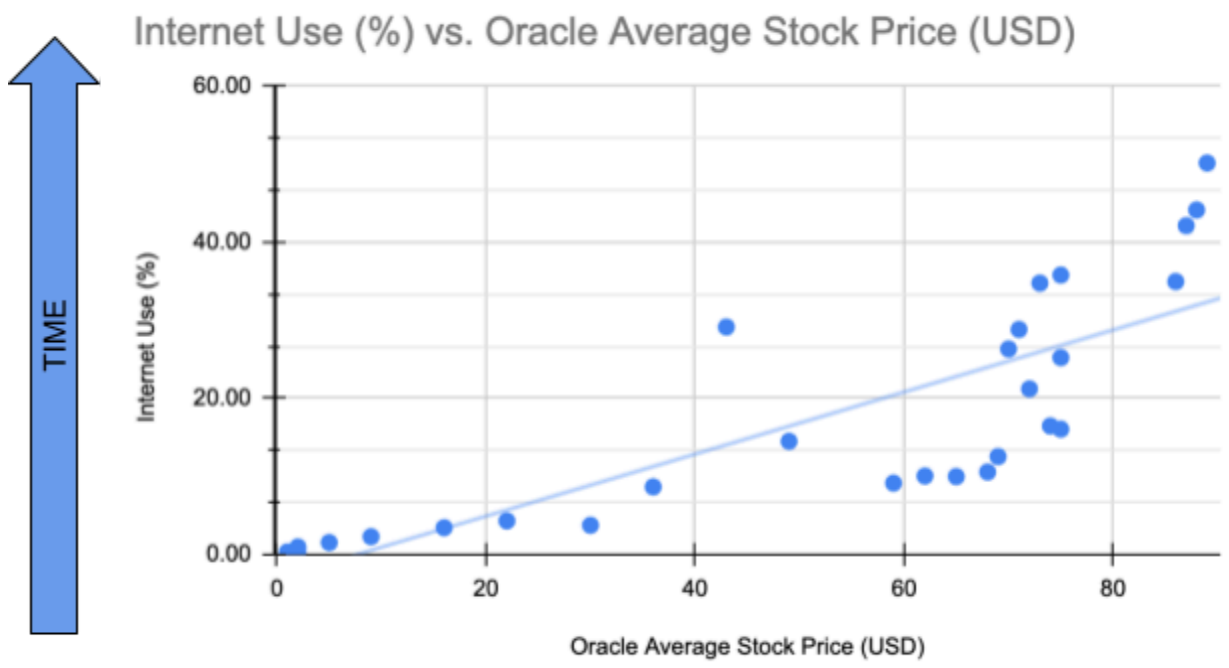
Is there a correlation between Internet Use (%) and Median Household Income (USD)?	X: Median Household Income (USD) Y: Internet Use (%) n = 29	$0.75 < r < 1$ $r = 0.7841$ $r^2 = 0.6148$	This is a strong positive correlation with the r-value being relatively close to 1, which means that high Internet Use (%) scores go with high Median Household Income scores (and vice versa). The r^2 value explains that approximately 61% of the variance of the Internet Use (%) being studied is explained by the variance of the Median Household Income (USD).



Is there a correlation between Internet Use (%) and Apple Average Stock Price (USD)?	X: Apple Average Stock Price (USD) Y: Internet Use (%) n = 29	$0.75 < r < 1$ $r = 0.6497$ $r^2 = 0.4221$	This is a moderate positive correlation with the r-value being greater by 0.5 (though not by much), which means there is a tendency for high Internet Use (%) scores to go with high Apple Average Stock Price scores (and vice versa). The r^2 value explains that approximately 42% of the variance of the Internet Use (%) being studied is explained by the variance of the Apple Average Stock Price (USD).



Is there a correlation between Internet Use (%) and Microsoft Average Stock Price (USD)?	X: Microsoft Average Stock Price (USD) Y: Internet Use (%) n = 29	$0.75 < r < 1$ $r = 0.6833$ $r^2 = 0.4669$	This is a moderate positive correlation with the r-value being greater than 0.5 (though not by much), which means there is a tendency for high Internet Use (%) scores to go with high Microsoft Average Stock Price scores (and vice versa). The r^2 value explains that approximately 47% of the variance of the Internet Use (%) being studied is explained by the variance of the Microsoft Average Stock Price (USD).



Is there a correlation between Internet Use (%) and Oracle Average Stock Price (USD)?	X: Oracle Average Stock Price (USD) Y: Internet Use (%) n = 29	$0.75 < r < 1$ $r = 0.8202$ $r^2 = 0.6727$	This is a strong positive correlation with the r-value being relatively close to 1, which means that high Internet Use (%) scores go with high Oracle Average Stock Price scores (and vice versa). The r^2 value explains that approximately 67% of the variance of the Internet Use (%) being studied is explained by the variance of the Oracle Average Stock Price (USD).

Discussion Pt.2

By removing the data values for the years 2008 and 2020 from the sample dataset, it is found that the overall strength of each correlation changed by a varying amount (when comparing r-values for each correlation from the original and modified datasets).

Certain variables, when related to Internet Use (%), grow stronger in the strength of their correlation (as in became closer to -1 or 1). This is the case for the correlations of the

Unemployment Rate (%), Apple Average Stock Price (USD), and Microsoft Average Stock Price (USD) with Internet Use (%), each of which experiences r-value changes of -0.1201, +0.0617, and +0.0694 respectively. However, despite becoming stronger in correlational strength, none of these updated r-values break the threshold of $-1 < r < -0.75$ or $0.75 < r < 1$ for the variables to be considered very strong in correlational strength.

Other variables, like Oracle Average Stock Price (USD), Trade Balance (millions USD), and Median Household Income (USD), witness a marginal change in their correlational coefficient (+0.0028, -0.0059, and -0.0003 respectively). This signals that the data values from 2008 and 2020 do not have much of an impact on skewing the strength of the correlations for the Oracle Average Stock Price, U.S. Trade Balance, and Median Household Income when related to U.S. Internet Use (%) for the years 1990-2020. Despite achieving marginal changes in their correlational coefficient, these variables actually grow stronger in correlational strength, solidifying that they are very strong in correlational strength when related to Internet Use (%).

The last variable, GDP Growth Percentage, sees an r-value change of +0.095 from the original correlational coefficient of -0.3416 to the updated coefficient of -0.2466. This change highlights an unusual weakening of the strength of the correlation, which is seemingly unexpected considering two of the negative GDP Growth Percentage data values from 2008 and 2020 are omitted from the dataset. This weakening of the correlation thus suggests that the correlation between GDP Growth Percentage and Internet Use (%) could be influenced by other factors and confounding variables. For this reason, it is essential for further research to be conducted into this specific relationship to identify any potential outliers, anomalies, biases, or historical fallacies that may result in this unexpected result.

Limitations Pt.2

These modified analyses hold limitations of their own concerning the dynamics of the data omissions. Firstly, these analyses test only 29 data values as opposed to the previous 31 values, with n=30 being the suggested sample size for a study to elicit accurate and generalizable conclusions. While the number of data values being tested is fewer than 30, it is close enough to the suggested sample size to be considered roughly statistically significant and acceptable for drawing conclusions, especially considering that this study involves the specific time span of 1990-2020 that cannot be expanded to before 1990 (since the software companies had not entered the stock market before then).

The modified analyses also pose the limitation of the recession years being limited to only a single year. In reality, the effects of these regressions did not last one year, with the Great Regression beginning in December 2007 and lasting until June 2009 and the COVID-19 Recession beginning in February 2020 and lasting until April 2020. However, for the sake of this study and to retain the sample size close to the suggested n=30, the effects of these two regressions were condensed to just one year's worth of data. Specifically, for the Great Recession from 2007-2009, the brunt of the effects came forth in 2008—hence the omission of 2008 data. For the COVID-19 Recession, despite lasting only two months and being the shortest economic recession in history, the effects of the economic downturn were so drastic that they warrant the omission of the data values for the entire year of 2020. These omissions thus produce the 29-value dataset for which additional correlational analyses are conducted.

Consequences and Implications

Given that the correlational test of each software company's average stock price with U.S. Internet Use (%) elicits positive and strong correlations, the assumption that American name-brand software companies supported U.S. Internet growth from 1990-2020 can be sustained. The other hypothesis tested in this study, that the growth of the U.S. economy supported U.S. Internet growth from 1990-2020, is not sustained due to weak and insignificant correlational results in two of the four tested economic indicators.

This study thus denies support to the claim that the U.S. economy has a stronger correlation to advancing the Internet than the software companies' contributions do. Rather, this study supports the claim that there exists a strong correlation between the success of American name-brand software companies and the growth of the Internet in the United States, with Internet Use (%) increasing as the Average Stock Price increases.

These findings highlight that there is a close relationship between software development and Internet access. They demonstrate that the growth of name-brand software companies, along with the increased innovation and spread of digital technologies that come as a result, have some relation to improving Internet access. These findings also present that there is not as much of a relationship between the economy and Internet access, suggesting that the growth of the Internet is not always related to an improving economy.

Based on this observation, it can be determined that the first step to improving Internet access in poorer nations is to not improve the economy of the nation, but rather to improve the technological spread and access of the nation. Rather than making efforts to revive and strengthen the economic system of a nation, the distribution of technology would prove more supportive of fostering a more interactive digital presence in that given nation. This is a

proposition that would require much more research, analysis, and modeling to support, but if supported, could revolutionize technological diffusion efforts for years to come and improve global connectivity as a whole.

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